CLINICAL STUDY REPORT

A Phase 3, randomized, double-blind, placebo-controlled, multicenter study to evaluate the efficacy and safety of XYZ-101 compared with placebo in patients with mild to moderate hypertension.

1. TITLE PAGE

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Study Title **A Phase 3, randomized, double-blind, placebo-controlled, multicenter study to evaluate the efficacy and safety of XYZ-101 compared with placebo in patients with mild to moderate hypertension.**	
Name of Test drug/ Investigational product:	XYZ-101
Indication:	Mild to Moderate hypertension
Protocol identification:	XYZ-101-HTN-301
Study Phase:	Phase 3
Study initiation date (first patient enrolled):	15 January 2025
Study completion date (last patient completed):	30 September 2025
Date:	15 October 2025
Version:	1.0

2. SYNOPSIS

Title of Study:	A Phase 3, randomized, double-blind, placebo-controlled, multicenter study to evaluate the efficacy and safety of XYZ-101 compared with placebo in patients with mild to moderate hypertension.
Study centre(s)	Multicentre
Studied period	First Patient Enrolled: 15 January 2025 Last Patient Completed: 30 September 2025
Study Phase	Phase 3
Objectives:	**Primary Objective** ➢ To evaluate the efficacy of XYZ-101 compared with placebo in reducing systolic blood pressure (SBP) from baseline to Week 12 in patients with mild to moderate hypertension. **Secondary Objectives** ➢ To evaluate the effect of XYZ-101 on diastolic blood pressure (DBP) from baseline to Week 12. ➢ To assess the proportion of patients achieving target blood pressure control (e.g., SBP <140 mmHg) at Week 12. ➢ To evaluate the safety and tolerability of XYZ-101 in the study population.
Methodology:	This was a Phase 3, randomized, double-blind, placebo-controlled, multicenter study conducted across approximately 15 clinical sites to evaluate the efficacy and safety of XYZ-101 in patients with mild to moderate hypertension. Approximately 300 patients were enrolled across multiple study centers. Patients were randomized in a 1:1 ratio to receive either XYZ-101 50 mg once daily or matching placebo for a duration of 12 weeks. The study consisted of a screening period of up to 2 weeks, a 12-week treatment period, and a follow-up visit.
Number of patients (planned and analysed):	A total of 300 patients were planned for enrollment in the study. Of these, 300 patients were randomized to receive either XYZ-101 (n=150) or placebo (n=150).

	The intent-to-treat (ITT) population included all randomized patients (n=300) and was used for efficacy analyses. The safety population included all patients who received at least one dose of study medication (n=296). A total of 4 randomized patients did not receive study treatment and were excluded from the safety population
Diagnosis and main criteria for inclusion and exclusion:	**Inclusion** Patients included in the study were adults diagnosed with mild to moderate hypertension. The main criteria for inclusion were: • Male or female patients aged 18 to 65 years • Seated systolic blood pressure (SBP) between 140 and 160 mmHg, inclusive • Seated diastolic blood pressure (DBP) between 90 and 100 mmHg, inclusive • Ability to provide written informed consent • Willingness to comply with study procedures **Exclusion** Patients were excluded from the study if they met any of the following criteria: • Severe hypertension (systolic blood pressure >160 mmHg or diastolic blood pressure >100 mmHg) • Secondary hypertension (e.g., due to renal, endocrine, or vascular causes) • History of significant cardiovascular events (e.g., myocardial infarction, stroke) within the past 6 months • Clinically significant hepatic or renal impairment • Known hypersensitivity to the investigational product or its components • Pregnant or lactating women • Use of prohibited concomitant medications that could interfere with study outcomes • Participation in another clinical trial within 30 days prior to screening.
Test product, dose and mode of administration	The investigational product, XYZ-101, was

	administered orally at a dose of 50 mg once daily. Patients randomized to the control group received a matching placebo, administered orally once daily. The study medication was supplied in tablet form and was taken with water, with or without food, throughout the 12-week treatment period
Duration of treatment:	The duration of treatment was 12 weeks. Patients were administered XYZ-101 50 mg or matching placebo orally once daily from Baseline (Day 0) through Week 12.
Reference therapy, dose and mode of administration:	The reference therapy consisted of a matching placebo, administered orally once daily. The placebo tablets were identical in appearance, packaging, and labeling to XYZ-101 to ensure maintenance of the double-blind design throughout the study.
Criteria for evaluation: **Efficacy:** **Safety:**	**Efficacy** **Primary Efficacy Endpoint** ➢ Change from baseline in systolic blood pressure (SBP) at Week 12 **Secondary Efficacy Endpoints** ➢ Change from baseline in diastolic blood pressure (DBP) at Week 12 ➢ Proportion of patients achieving target blood pressure control (SBP <140 mmHg) at Week 12 ➢ Blood pressure measurements were obtained at each scheduled study visit using standardized procedures. **Safety** Safety was evaluated throughout the study based on: ➢ Adverse events (AEs) ➢ Vital signs ➢ Laboratory parameters ➢ Physical examinations ➢ Adverse events were graded according to CTCAE v5.0 and assessed for severity and relationship to the study treatment.
Statistical methods:	Efficacy analyses were performed on the intent-to-treat (ITT) population, which included all randomized patients. Safety analyses were conducted on the safety population, defined as all patients who received at least one dose of study

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<td>medication.

Continuous variables, including systolic and diastolic blood pressure, were summarized using descriptive statistics (mean, standard deviation, median, and range). Changes from baseline were analyzed using appropriate statistical methods, and comparisons between treatment groups were performed.

Categorical variables, including responder rates, were summarized using frequencies and percentages. Comparisons between groups were conducted using appropriate statistical tests.

All statistical tests were two-sided, and a p-value of <0.05 was considered statistically significant.</td>
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<tr>
<td colspan="2">SUMMARY - CONCLUSIONS

EFFICACY RESULTS:

Treatment with XYZ-101 resulted in a clinically meaningful and statistically significant reduction in systolic blood pressure compared with placebo at Week 12 (−16 mmHg vs −6 mmHg; 95% CI: −13.4 to −6.6; $p<0.001$). Improvements were also observed in diastolic blood pressure and in the proportion of patients achieving target blood pressure control (65% vs 30%), supporting the overall efficacy of XYZ-101.

SAFETY RESULTS:

XYZ-101 was generally well tolerated, with most adverse events being mild to moderate in severity, a low incidence of treatment discontinuations, and no new safety signals identified.

CONCLUSION:

Overall, the results of this Phase 3 study demonstrate that XYZ-101 provides effective blood pressure reduction with a favorable safety profile in patients with mild to moderate hypertension.</td>
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3. TABLE OF CONTENTS

4. LIST OF ABBREVIATIONS

ACE inhibitors	Angiotensin-converting enzyme inhibitors
AEs	Adverse events
ARBs	Angiotensin receptor blockers
CI	Confidence Interval
CVD	Cardiovascular disease
DBP	Diastolic blood pressure
ECG	Electrocardiogram
HTN	Hypertension
SBP	Systolic blood pressure
ITT	Intent-to-treat

STUDY CALENDAR

Assessment	Screening	Baseline	Wk 2	Wk 4	Wk 8	Wk 12	Follow-up
Informed Consent	X						
Inclusion/Exclusion Criteria	X						
Demographics		X					
Body Weight		X				X	
Medical History	X						
Medication History	X						
Physical Examination	X	X		X		X	
Blood Pressure (SBP/DBP)	X	X	X	X	X	X	X
Laboratory Tests	X			X		X	
ECG	X					X	
Pulse Rate	X	X	X	X	X	X	X
Pregnancy Test	X					X	
Concomitant Medications	X	X	X	X	X	X	X
Study Drug Administration		X	X	X	X	X	
Adverse Events			X	X	X	X	X
Drug Accountability				X		X	

5. INTRODUCTION

Hypertension (HTN) is a chronic medical condition characterized by persistent elevation in arterial blood pressure. According to current guidelines, hypertension is defined as systolic blood pressure (SBP) ≥130 mmHg and/or diastolic blood pressure (DBP) ≥80 mmHg. Hypertension is widely recognized as a major modifiable risk factor for cardiovascular disease (CVD) and all-cause mortality worldwide.[1]

Globally, hypertension represents a significant public health burden, with an estimated prevalence of approximately 31% among adults, corresponding to over 1 billion individuals. The burden of hypertension continues to rise, driven by population ageing and increased exposure to lifestyle-related risk factors such as high sodium intake, low physical activity, and unhealthy dietary patterns.[2]

Large-scale epidemiological analyses have demonstrated that mean blood pressure levels have remained relatively stable globally over the past several decades, although regional variations persist. Despite this, the absolute number of individuals affected by hypertension has increased substantially due to population growth and ageing, underscoring the ongoing need for effective management strategies.[3]

Pharmacological management of hypertension includes several classes of antihypertensive agents, such as angiotensin-converting enzyme inhibitors (ACE inhibitors), angiotensin receptor blockers (ARBs), calcium channel blockers (CCBs), diuretics, and beta-blockers. Treatment selection is guided by patient-specific factors including age, comorbidities, and overall cardiovascular risk, as outlined in major international guidelines.[4]

Despite the availability of multiple antihypertensive therapies, a substantial proportion of patients fail to achieve optimal blood pressure control, highlighting an unmet medical need for effective and well-tolerated treatments. Therefore, this Phase 3 study was conducted to evaluate the efficacy and safety of XYZ-101 in patients with mild to moderate hypertension.

6. STUDY OBJECTIVES

Primary Objective

- To evaluate the efficacy of XYZ-101 compared with placebo in reducing systolic blood pressure (SBP) from baseline to Week 12 in patients with mild to moderate hypertension.

Secondary Objectives

- To evaluate the effect of XYZ-101 on diastolic blood pressure (DBP) from baseline to Week 12.
- To assess the proportion of patients achieving target blood pressure control (e.g., SBP <140 mmHg) at Week 12.
- To evaluate the safety and tolerability of XYZ-101 in the study population.

7. INVESTIGATIONAL PLAN

7.1 Overall Study Design

This was a Phase 3, randomized, double-blind, placebo-controlled, multicenter study conducted across approximately 15 clinical sites to evaluate the efficacy and safety of XYZ-101 in patients with mild to moderate hypertension.

Approximately 300 patients were enrolled and randomized in a 1:1 ratio to receive either XYZ-101 50 mg once daily or matching placebo for a duration of 12 weeks. The study consisted of a screening period of up to 2 weeks, a 12-week treatment period, and a follow-up visit.

7.2 Study Population

Eligible patients were adults aged 18 to 65 years with a diagnosis of mild to moderate hypertension, defined by seated systolic blood pressure between 140 and 160 mmHg and diastolic blood pressure between 90 and 100 mmHg.

Key exclusion criteria included severe hypertension, significant cardiovascular disease, and use of prohibited concomitant medications that could interfere with study outcomes.

7.3 Treatments

Patients received either XYZ-101 50 mg or matching placebo administered orally once daily. Treatment compliance was monitored throughout the study using drug accountability records.

7.4 Randomization and Blinding

Patients were randomized using a centralized randomization schedule. The study was conducted in a double-blind manner, with both patients and investigators unaware of treatment allocation.

7.5 Efficacy and Safety Assessments

Efficacy was primarily assessed by the change in systolic blood pressure from baseline to Week 12. Secondary endpoints included changes in diastolic blood pressure and the proportion of patients achieving target blood pressure control.

Safety was evaluated through monitoring of adverse events, vital signs, laboratory parameters, and physical examinations throughout the study

8. CRITERIA FOR EVALUATION:

Efficacy

Primary Efficacy Endpoint

- Change from baseline in systolic blood pressure (SBP) at Week 12

Secondary Efficacy Endpoints

- Change from baseline in diastolic blood pressure (DBP) at Week 12
- Proportion of patients achieving target blood pressure control (SBP <140 mmHg) at Week 12
- Blood pressure measurements were obtained at each scheduled study visit using standardized procedures.

Safety

Safety was evaluated throughout the study based on:

- Adverse events (AEs)
- Vital signs
- Laboratory parameters
- Physical examinations
- Adverse events were graded according to CTCAE v5.0 and assessed for severity and relationship to the study treatment.

9. STATISTICAL METHODS

Efficacy analyses were performed on the intent-to-treat (ITT) population, which included all randomized patients. Safety analyses were conducted on the safety population, defined as all patients who received at least one dose of study medication.

Continuous variables, including systolic and diastolic blood pressure, were summarized using descriptive statistics (mean, standard deviation, median, and range). Changes from baseline were analyzed using appropriate statistical methods, and comparisons between treatment groups were performed.

Categorical variables, including responder rates, were summarized using frequencies and percentages. Comparisons between groups were conducted using appropriate statistical tests.

All statistical tests were two-sided, and a p-value of <0.05 was considered statistically significant.

10. EFFICACY EVALUATION

10.1 Analysis Populations

The efficacy analysis was performed on the intent-to-treat (ITT) population, which included all randomized patients (n=300).

10.2 Demographic and Baseline Characteristics

The demographic and baseline characteristics were comparable between the treatment groups.

Table 1: Demographic and Baseline Characteristics

Parameter	XYZ-101 (n=150)	Placebo (n=150)
Age (years), mean (SD)	52 (10.2)	51 (9.8)
Male, n (%)	88 (58.7%)	85 (56.7%)
Female, n (%)	62 (41.3%)	65 (43.3%)
SBP (mmHg), mean (SD)	148 (6.5)	147 (6.8)
DBP (mmHg), mean (SD)	94 (4.2)	93 (4.5)

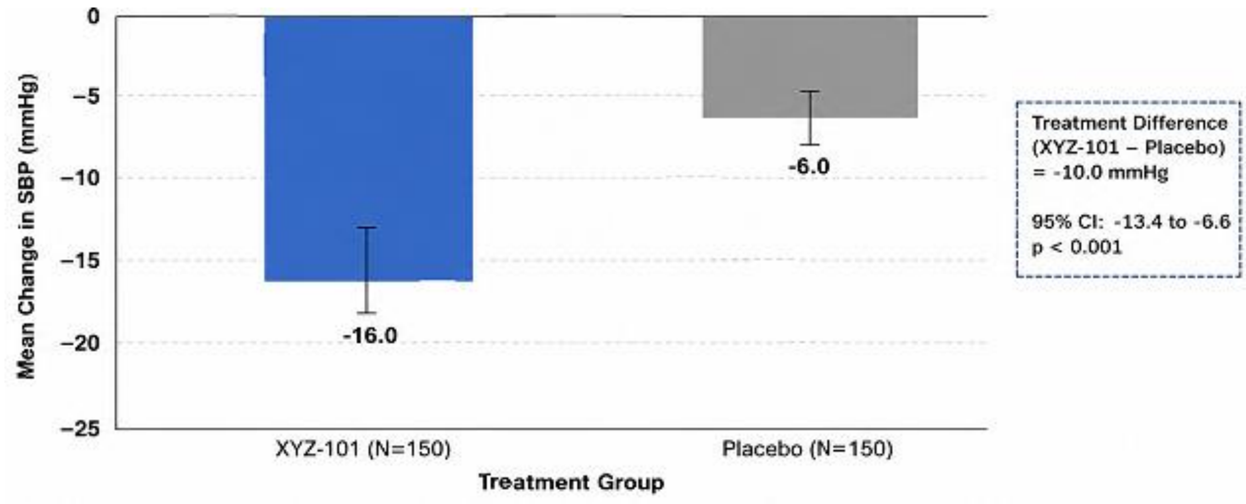
10.3 Efficacy Results

Primary Endpoint

Treatment with XYZ-101 resulted in a greater reduction in systolic blood pressure compared with placebo at Week 12.

Table 2: Change in SBP from Baseline to Week 12

Parameter	XYZ-101	Placebo	Treatment Difference	95% CI	p-value
Mean change in SBP (mmHg)	−16	−6	−10	−13.4 to −6.6	<0.001



Note: Bars represent mean change from baseline to Week 12. Error bars represent ± SD. Abbreviations: CI = Confidence Interval; SBP Systolic Blood Pressure; SD = Standard Deviation.

Figure 1: Change in SBP from Baseline to Week 12

Secondary Endpoint

XYZ-101 also demonstrated improvements in diastolic blood pressure and responder rates compared with placebo.

Table 3: Change in DBP from Baseline to Week 12

Parameter	XYZ-101	Placebo	Treatment Difference	p-value
Mean change in DBP (mmHg)	−10	−4	−6	<0.001
Responders (SBP <140 mmHg), %	65%	30%	+35%	<0.001

10.4 Interpretation of Efficacy Results

As shown in Table 2, treatment with XYZ-101 resulted in a substantial reduction in systolic blood pressure from baseline to Week 12 compared with placebo. Patients receiving XYZ-101 demonstrated a mean reduction of 16 mmHg, whereas those receiving placebo showed a reduction of 6 mmHg.

The between-group difference of 10 mmHg was statistically significant (95% CI: −13.4 to −6.6; $p<0.001$), indicating a clear treatment effect of XYZ-101. This reduction is considered clinically meaningful and supports the efficacy of XYZ-101 in improving blood pressure control in patients with mild to moderate hypertension.

Treatment with XYZ-101 resulted in a greater reduction in diastolic blood pressure compared with placebo (−10 mmHg vs −4 mmHg), corresponding to a treatment difference of −6 mmHg.

A higher proportion of patients in the XYZ-101 group achieved target systolic blood pressure control compared with placebo (65% vs 30%), representing an absolute difference of 35%.

11. SAFETY EVALUATION

11.1 Safety Population

The safety analysis was performed on all patients who received at least one dose of study medication (safety population, n=296).

A total of 4 randomized patients did not receive study treatment and were excluded from the safety population.

11.2 Overview of Adverse Events

Overall, XYZ-101 was well tolerated. The majority of adverse events (AEs) were mild to moderate in severity, and no clinically significant differences were observed between the treatment groups.

11.3 Summary of Adverse Events

Table 4: Summary of Adverse Events

Parameter	XYZ-101 (n=148)	Placebo (n=148)
Patients with ≥1 AE, n (%)	45 (30.4%)	38 (25.7%)
Patients with serious AEs, n (%)	2 (1.4%)	2 (1.4%)
Patients with drug-related AEs, n (%)	20 (13.5%)	12 (8.1%)
Discontinuations due to AEs, n (%)	3 (2.0%)	2 (1.4%)

11.4 Common Adverse Events

Table 5: Common Adverse Events (≥5% of patients)

Adverse Event	XYZ-101 (%)	Placebo (%)
Headache	12%	8%
Dizziness	8%	5%
Nausea	5%	3%

11.5 Other Safety Findings

No clinically significant abnormalities were observed in laboratory parameters, vital signs, or ECG assessments. There were no deaths reported during the study.

11.6 Interpretation of Safety Results

XYZ-101 was generally well tolerated in patients with mild to moderate hypertension. The incidence of adverse events was slightly higher in the XYZ-101 group compared with placebo; however, most events were mild in severity and did not lead to treatment discontinuation.

No new or unexpected safety signals were identified during the study.

STUDY LIMITATIONS

This study has certain limitations. The treatment duration of 12 weeks may not fully capture long-term efficacy and safety outcomes. Additionally, the study population was limited to patients with mild to moderate hypertension, which may limit the applicability of the findings to more severe populations. The absence of an active comparator also restricts comparative effectiveness assessment.

12. DISCUSSION AND OVERALL CONCLUSIONS

In this Phase 3 study, treatment with XYZ-101 demonstrated a clinically meaningful and statistically significant reduction in systolic blood pressure compared with placebo. Improvements were also observed in diastolic blood pressure and responder rates.

The observed reduction in systolic blood pressure is considered clinically meaningful and may translate into a reduction in cardiovascular risk.

The safety profile of XYZ-101 was favorable, with most adverse events being mild to moderate in severity. No unexpected safety concerns were identified.

Although limited by its short duration and controlled setting, the study found that XYZ-101 was effective and well tolerated in mild to moderate hypertension, with an overall favorable benefit–risk profile.

13. LIST OF REFERENCES

[1] Whelton PK, Carey RM, Aronow WS, et, al.2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Hypertension. 2018 Jun;71(6):1269-1324. doi: 10.1161/HYP.0000000000000066.

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